Not All Pairs Need All Correspondences: Certified Adaptive Budgets for Robust Point Cloud Registration

Supplementary material

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A. Certification: statements and proofs

Learn-then-Test with fixed-sequence testing. Let $\lambda_1 \prec \lambda_2 \prec \cdots \prec \lambda_L$ be the parameters of a policy family, ordered from conservative to aggressive, and let $\hat{R}_n(\lambda)$ be the empirical risk of the policy on n calibration pairs (the fraction of pairs registered by the reference but not by the policy). For each λ_l we test $H_l : R(\lambda_l) > \varepsilon$ with the exact binomial p-value $p_l = \Pr\big(\mathrm{Bin}(n, \varepsilon) \le n\hat{R}_n(\lambda_l)\big)$, and return $\hat{\lambda} = \lambda_{l^*}$ with $l^* = \max\{l : p_1, \ldots, p_l \le \delta\}$ (no policy if $p_1 > \delta$). Because the order is fixed before seeing the data, testing in sequence controls the family-wise error rate at δ, so $\Pr\big(R(\hat{\lambda}) \le \varepsilon\big) \ge 1 - \delta$ when calibration and test pairs are exchangeable [1]. With no loss among the n calibration pairs, $p_1 = (1-\varepsilon)^n$, so no level-ε certificate exists unless $n \ge \log\delta / \log(1-\varepsilon)$: 230 pairs for $\varepsilon = 1\,\%$ and $\delta = 0.1$, and 339 per group at level $\delta/3$.

Proof of Proposition 1. Let the groups $g = 1, \ldots, G$ be defined by an observable $g(z_1)$ of the first stage, and let $\hat{\lambda}_g$ be returned by the procedure above on the calibration pairs of group g at level δ/G. Define the events $\mathcal{E}_g = \{R_g(\hat{\lambda}_g) \le \varepsilon\}$, where R_g is the risk on test pairs of group g. By exchangeability within groups, $\Pr(\mathcal{E}_g^c) \le \delta/G$, so by the union bound $\Pr\big(\cap_g \mathcal{E}_g\big) \ge 1 - \delta$. On this event, for a test distribution with arbitrary group proportions $w_g \ge 0$, $\sum_g w_g = 1$, the risk of the combined policy is $R = \sum_g w_g R_g(\hat{\lambda}_g) \le \varepsilon$. □

The marginal certificate needs exchangeability of the whole calibration and test distributions, which fails when the test set has a different mix of easy and hard pairs (*e.g.* the KITTI raw drives, whose full-budget recall is 61.8 % against 92.3 % on the odometry pairs that calibrate them). The conditional certificate only needs exchangeability within each group.

Choosing the candidate. The statements above hold for one candidate (ladder, policy family and its fixed sequence) and one taxonomy (group edges) that do not depend on the calibration pairs. We choose both on the fitting pairs only, and use nothing of the calibration set but its size, which fixes how many losses a certificate can accept; conditionally on the fitting pairs, the chosen candidate is fixed and the guarantee applies to it unchanged. Choosing instead the cheapest of L candidates certified on the same calibration pairs only guarantees $1 - L\delta$ by the union bound, and the cheapest is the one whose certificate most likely erred on the aggressive side.

Truncated ladders. If the last budget N_K of a ladder is below the number of correspondences, even the policy that never stops early has a positive risk. We prepend this always-continue policy to every fixed sequence as λ_0; the sequence is still fixed in advance, so the guarantee above is unchanged, and a ladder whose H_0 is not rejected (in some group, for the conditional certificate) yields no certified policy.

B. Implementation and protocol

Protocol v2. All splits, rules and configurations of the paper's experiments were pre-registered (file `PROTOCOL.md` of the code release) before the new data were downloaded or processed. *Indoor*: the 75 3DMatch training scenes are permuted with seed 20261019 and assigned to the blind test until it holds at least 3,500 pairs with overlap above 10 % (9 scenes), then to calibration until it holds as many (13 scenes); the rest are fitting scenes. The blind test is a random sample of 3,000 of its pairs, reported above and below 30 % overlap; calibration is a random sample of 3,000 pairs, and the 1,000 fitting pairs are the earlier training sample restricted to fitting scenes, topped up at random. The 3DMatch validation scenes, which served the design, are not used with FPFH. *Outdoor*: see the KITTI paragraph below. *Deviation*: the raw drives outside odometry sequences 0–10 yield 1,332 pairs under FCGF's rule, not the 2,400 the outdoor split assumed; before any of their scans was downloaded, the split was amended so that FPFH is certified on odometry pairs and tested on all raw-drive pairs, and FCGF is certified on 42 drives and tested on the other 7. Each blind test set was evaluated once, after every selection had been written to disk.

Correspondences. FPFH [7] on 5 cm voxels (indoors) or 30 cm voxels (outdoors), with normal and feature radii of 2 and 5 voxels. For each source point we take the nearest target descriptor, sort the matches by Lowe ratio and keep at

most 4,000.

Estimators. SC^2-PCR and TurboReg are re-implemented in NumPy following the papers and the official code. Table 1 compares their full-budget recall with the published numbers. With FCGF features, the input closest to the published runs, both come within 0.5–1.3 pp of the published recall on 3DMatch and 3DLoMatch, so most of the 2.0–4.4 pp gap with FPFH likely lies in our FPFH correspondences rather than in the estimators; on KITTI the gap remains 3.3–4.3 pp with FCGF, where our estimator thresholds were chosen on FPFH validation pairs. Every speed-up in the paper compares an estimator with itself, so implementation speed cancels out to first order.

Table 1. **Re-implementations vs. published recall** (%, full budget, same success criteria). FPFH: our correspondences, test sets. FCGF: features of our re-implementation of the network (identical to those released with SC^2-PCR), all test pairs. Published values from TurboReg [10] (TurboReg with $K_1 = 2{,}000$ indoors and 500 pivots on KITTI).

	3DMatch		3DLoMatch		KITTI	
Estimator	ours	publ.	ours	publ.	ours	publ.
SC^2-PCR, FPFH	81.5	83.7	36.6	38.6	92.3	96.4
TurboReg, FPFH	81.9	84.1	36.6	41.0	92.3	97.8
SC^2-PCR, FCGF	92.5	93.2	58.2	58.7	93.9	97.1
TurboReg, FCGF	92.6	93.6	58.4	59.7	93.9	98.2

KITTI odometry (development). We follow the FCGF/Predator/GeoTransformer protocol [2, 4, 6] and use GeoTransformer's pairs and ICP-refined ground truth: sequences 0–5 for training, 6–7 for validation and 8–10 for testing (555 test pairs). The official velodyne archive (85 GB) supports HTTP range requests, so we read its index remotely and download only the scans used by these pairs, voxelised at 0.3 m; 300 training pairs fit the controller, and the other 1,058 training pairs and the 180 validation pairs certify it. A pair is registered if RE $\leq 5^\circ$ and TE ≤ 60 cm. Nearest descriptors are found by brute force outdoors (a k-d tree degrades on its many near-identical ground descriptors, taking minutes per pair); on 3DMatch pairs the two searches agree on 99.9 % of the nearest neighbours. Distance thresholds scale with the inlier threshold (0.6 m vs. 0.1 m indoors); the compatibility thresholds of the two estimators were chosen by full-budget recall on the validation sequences (Tab. 2), and the ordering uses SC^2-PCR's threshold.

KITTI raw drives (blind). The KITTI raw recordings contain 145 drives that belong to none of the odometry sequences 0–10, so neither FCGF nor this work had used them; 49 of them move far enough to form pairs. Scans are fetched by HTTP range requests from the official archives. Pairs follow FCGF's rule: start at the first frame, take as partner the frame before the first one more than 10 m away, and start again after the partner, which gives 1,332 pairs. Ground truth comes from the drive's GPS/IMU poses refined by point-to-point ICP exactly as in FCGF (5 cm voxels, 0.2 m correspondence distance, 200 iterations); 22 pairs on which ICP moved the pose by more than 1 m or 5° (a failed refinement, by the pre-registered rule) were dropped, leaving 1,310 pairs. FPFH is tested on all of them; FCGF is certified on the 1,107 pairs of 42 drives and tested on the 203 pairs of the other 7, assigned by a seeded permutation of the drives.

Table 2. **KITTI compatibility thresholds**, chosen by full-budget recall on the validation sequences (180 pairs).

Estimator	threshold (m)	val. recall (%)	time (ms)
SC^2-PCR (d)	0.300	96.1	5,018
	0.600	96.1	4,867
	1.200 (chosen)	96.1	4,855
TurboReg (τ)	0.072 (chosen)	96.1	6,028
	0.180	96.1	6,034
	0.360	96.1	6,106
	0.720	93.9	6,248

FCGF. FCGF [2] features exist only for the test sets, so we re-implemented the network (a sparse residual U-Net) in plain PyTorch, with MinkowskiEngine's quantisation and kernel ordering, and load the official weights (3DMatch, 5 cm; KITTI, 30 cm, 32 dimensions) with a restricted unpickler. Run on the input points of the released test features (8 3DMatch fragments and 4 KITTI scans, every keypoint), it reproduces them with a minimum cosine similarity of 1.00000 and an absolute difference of at most 1.2×10^{-6} (`results/protocol_v2/fcgf_verification.txt` in the code release), so the FCGF results do not depend on the re-implementation. With it we compute FCGF features for every split. FCGF was trained on all 3DMatch training scenes and on odometry sequences 0–5: indoors it is fitted on the 1,000 fitting pairs, certified on the 1,331 pairs of the 3DMatch validation scenes (not used in FCGF's training) and tested on the development benchmarks; outdoors it is fitted on the 300 odometry training pairs and certified and tested on the raw drives. Correspondences, budgets and estimator thresholds are those of the FPFH experiments. On the odometry training pairs nearly every first stage of FCGF succeeds, so the outdoor controller sees almost no failure; the certificate then rejects every early stop on the calibration drives, and CAB runs the full budget (Sec. 4.3 of the main paper).

Controller. Features (per stage): $\log(1 + \text{inliers})$, inlier fraction of the budget, fraction of all correspondences and of the 250 most distinctive explained by the estimate (inlier threshold of the dataset), $\log(1 + \text{pose change})$ in degrees and metres and an agreement flag ($< 5^\circ$, < 0.2 m), the stage index, $\log N$, the median Lowe ratio of the budget, and the gain in explained correspondences. p_{now} and p_{fin} are logistic regressions (standardised features, $C = 1$)

fitted on all stages of the training pairs (indoors: the 1,000 fitting pairs; KITTI: 300 odometry training pairs; FCGF: the same pairs with FCGF features). Threshold grids: $\lambda \in \{1.01, 0.995, \ldots, 0.5\}$ (18 values), $\gamma \in \{0, 0.02, 0.05, 0.1, 0.15, 0.2, 0.3\}$, and 42 cost prices for the Markov controller. Difficulty groups are the terciles, on the fitting pairs, of $\log(1+\text{explained correspondences})$ after the first stage. Candidate ladders: nine that end at the full budget (from 250 or 500, through subsets of 1,000 and 2,000; among them the doubling and $\times 4$ ladders), four that end at 2,000, and the single budgets 1,000 and 2,000 (15 ladders, six of which end below 4,000), plus the full budget alone, which needs no certificate. *Choice* (Sec. 3.4 of the main paper): the fitting pairs are split into five folds, each scored by a controller fitted on the other four; for every candidate (ladder and family; for give-up also γ), the most aggressive parameters of its fixed sequence with at most k losses on these pairs are taken, where k/n is the largest loss share the exact binomial test accepts for the calibration set's size n (no loss per group for the group-conditional certificate with our pools, about $0.6\,\%$ for the marginal one), and the candidate with the lowest mean time is certified.

Cost accounting and timing. Each distinct computation is charged once: the ordering when the first truncated budget is visited, and a budget that covers all correspondences only once even if several ladder stages coincide. *Operation counts* are hardware-independent: each run is charged the dominant arithmetic of its steps on its N correspondences. For SC^2-PCR: N^2 compatibility tests, $10N^2$ for the power iterations of the leading eigenvector, SN^2 for the second-order compatibility of the S seeds, $k_1^3 + k_2^3$ per seed for local consensus, SN for scoring the hypotheses and N per refinement iteration. For TurboReg: N^2 tests and N^3 for the second-order graph, $K_1 N$ for the third vertex of the K_1 pivot edges, N per 3-clique hypothesis and per refinement iteration. For our ordering: P^2 tests among the $P = 500$ most distinctive correspondences, then $Nm + Nm^2$ for the $m = 100$ anchors. The ratio of the mean count of the full budget to that of a configuration, over the full test set, is the operation reduction of Tables 1 and 2 of the main paper. The counts ignore fixed per-run overheads and data-dependent work that leaves the leading terms unchanged, such as TurboReg's fuller clique search on consistent input (Sec. 4.6 of the main paper), so they complement rather than replace wall-clock time. Offline costs come from per-pair logs of every stage, timed single-threaded with six parallel workers; each method is compared with a reference timed in the same run. All wall-clock claims are confirmed live: one process with the default multi-threaded BLAS (OpenBLAS, 16 threads), the same random test pairs for every configuration, and the order of the runs rotated from pair to pair. Offline (single-threaded) and live (multi-threaded) speed-ups can differ, since multi-threading does not accelerate small and large budgets equally: $1.67\times$ offline vs. $1.56\times$ live for SC^2-PCR on the held-out scenes, $1.15\times$ vs. $1.08\times$ on KITTI odometry and $1.06\times$ vs. $1.04\times$ on FCGF 3DMatch; the paper therefore reports live speed-ups. Live and offline stopping decisions agreed on at least 97 % of the pairs in every live run. A pair is timed again when one of its runs takes more than ten times the median full-budget time of its test set, which no run can do legitimately (an adaptive run costs at most about two full runs); no pair of the protocol-v2 timing runs met it (0 re-timed pairs).

C. Protocol v3: the hardness gate and the fresh blind test

Why a second blind test. After the protocol-v2 blind evaluation, exploration on fitting, calibration and development data only (the blind sets were never loaded) found two improvements: choosing the controller's classifier on the fitting pairs, and a hardness gate. Because they were found after the v2 blind sets had been seen, the final method was tested again on fresh data under a second pre-registered protocol (file `PROTOCOL.md`, committed before the split was generated).

Data. The 53 fitting scenes of protocol v2 were permuted with seed 20261021 and assigned to the fresh blind test until it held at least 3,500 pairs with overlap above 10 % (10 scenes); 3,000 pairs were sampled at random. Before any of them was processed, a leakage check found that 377 had appeared in earlier logs as fitting or training pairs (never as test pairs: protocol-v2 fitting pairs, the first protocol's training sample and the pose-search training logs); they were removed (deviation 1 of protocol v3), leaving 2,623 pairs that no experiment had processed under any role, 1,976 above and 647 below 30 % overlap. The final method is fitted on 711 protocol-v2 fitting pairs outside these scenes and 289 new pairs of the remaining fitting scenes, and certified on the protocol-v2 calibration pairs. Every 3DMatch training scene had appeared in some earlier log, so scenes that are fresh with respect to all earlier work do not exist; the fresh scenes are disjoint from every set that fits, calibrates, chooses or gates the final method. No fresh outdoor or FCGF data remained: on the protocol-v2 sets the final method is reported as not blind.

Classifier choice. Four kinds are candidates for the controller's two classifiers: logistic regression ($C = 1$, the protocol-v2 controller, and $C = 0.1$), logistic regression on degree-2 features ($C = 0.1$) and gradient-boosted trees (depth 3, 300 iterations, learning rate 0.05). For each dataset, estimator and ordering, choose-then-certify (Sec. 3.4 of the main paper) runs once per kind on the fitting pairs, and the kind whose chosen candidate has the lowest out-of-fold cost is certified; the calibration pairs play no part in the choice. Chosen kinds (compatibility order, SC^2-PCR

/ TurboReg): fresh scenes logistic (C=0.1) / quadratic logistic; indoor FPFH boosted trees / boosted trees; KITTI FPFH quadratic logistic / logistic (C=0.1); indoor FCGF quadratic logistic / boosted trees (Tab. 5 lists every configuration).

Gate. Fourteen signals of the putative correspondences (Lowe-ratio order, cap 4,000), computed before any estimator run: $\log N$; the 10th and 50th percentiles of the Lowe ratio and its mean over the 250 most distinctive correspondences; the largest, 10th-largest and mean first-order degree of the 100 anchors among the 500 most distinctive correspondences (the ordering's first pass); the 1st, 10th, 50th and 250th largest second-order anchor support (Eq. 2 of the main paper); the number of correspondences consistent with at least 25 and 50 % of the anchors; and the anchors' mutual consistency. The label of a training pair is whether CAB, as certified, costs more than the full budget on it; logistic regression or boosted trees and a threshold are chosen by the 5-fold out-of-fold total cost on the fitting and calibration pairs. A gated pair runs the full budget, whose result is the reference, so the gate can only remove losses: if L is the set of pairs the certified policy loses, the gated policy loses L minus the gated pairs, overall and in every difficulty group, and every certificate holds unchanged. The gate's signals return the compatibility order as well, so a pair that is not gated reuses them; the order is identical to that of the ordering alone.

Cost accounting. The signals cost 11.5 ms per pair and replace the ordering's first pass (11.6 ms alone), and one prediction costs 0.6 ms (logistic) or 3.2 ms (trees), measured in one process with the default multi-threaded BLAS (`scripts/gate_cost.py`). Offline, every pair is charged one prediction, gated pairs the signals plus the full budget, and other pairs the difference between signals and ordering (compatibility order) or the signals (ratio order). The gate was trained, and first evaluated, with an earlier accounting that charged no prediction and took the signals' cost from the feature runs, which were timed under parallel load; the evaluation was then re-run with the consistent accounting above. It reproduced every choice and every per-pair outcome and changed the offline speed-ups by at most 0.02 for the group-conditional certificates and 0.11 for the marginal ones, in either direction.

Live overhead. Live, on the fresh pairs that the gate lets through and on which CAB with and without the gate stop at the same stage, so that the estimator does the same work, the gated method takes 40–48 ms longer per pair (median), an order of magnitude more than the measured cost of signals and prediction. The process holds two OpenBLAS thread pools and the OpenMP pool of the tree classifier, each sized to the 16 hardware threads; contention between them after a prediction is the likely cause, but we did not establish it, and the reported times include the overhead as measured. For SC^2-PCR above 30 % overlap it outweighs the gate's saving (Table 1 of the main paper).

Untimed configuration. With FCGF on KITTI the certified final method is the full budget itself, for both estimators and orderings, as in protocol v2; its live timing was stopped as redundant, and the tables report 1.00× by construction.

D. Additional results

Which pairs benefit. Table 8 groups pairs by the number of true correspondences among the 100 anchors and compares SC^2-PCR's success at $N = 250$ under both orderings.

Per-pair latency. CAB lowers the mean time per pair, not the worst case (Tab. 9). On 3DMatch most pairs stop after the first stage, so its median latency is a small fraction of the full budget's (251 vs. 727 ms for SC^2-PCR). A pair that needs the whole ladder also pays for the stages tried before, so on such pairs CAB is slower than one full run: its 99th-percentile latency is up to 36 % above the full budget's (TurboReg on KITTI raw), and on 3DLoMatch, where most pairs run the whole ladder, it is slower on 89–90 % of the pairs, by a few percent. An application with a hard per-pair deadline should therefore plan for the whole ladder.

Headroom. Table 10 compares CAB with two oracles on the same ladder (Sec. 4.5 of the main paper).

Ordering ablation. Table 11 isolates the choice of anchors on the validation scenes. One pass, with the most distinctive correspondences as anchors, triples the inliers in the cheap budget (82.6 vs. 27.8) but changes SC^2-PCR's success at $N = 250$ by only +1.5 pp and *lowers* it at 500 and 1,000 (-3.4 and -3.7 pp); two passes gain +7.5 and +1.4 pp at $N = 250$ and 500, the budgets at which the selected ladders start. Mean inlier yield is a misleading design target: it is dominated by easy pairs, while success is decided by the hard ones.

Estimator cost depends on its input. On consistency-ordered budgets, SC^2-PCR's first stage becomes cheaper (52 → 34 ms; fewer seeds survive non-maximum suppression), whereas TurboReg's becomes 4.3× more expensive at an unchanged operation count, because a consistent subgraph fills all of its candidate cliques. This is why the ordering buys TurboReg accuracy more than time, and why wall-clock time stays our primary cost measure [3].

E. Limitations revisited

Four questions about the protocol can be answered with the logs we already have: how many calibration pairs a certificate needs, how the guarantee holds group by group, how much the choice of ε and G matters, and what bounding the worst case costs. The studies that re-run the selection (calibration size, sensitivity, re-fitted controllers) use the development benchmarks only, so the blind test sets are evaluated once.

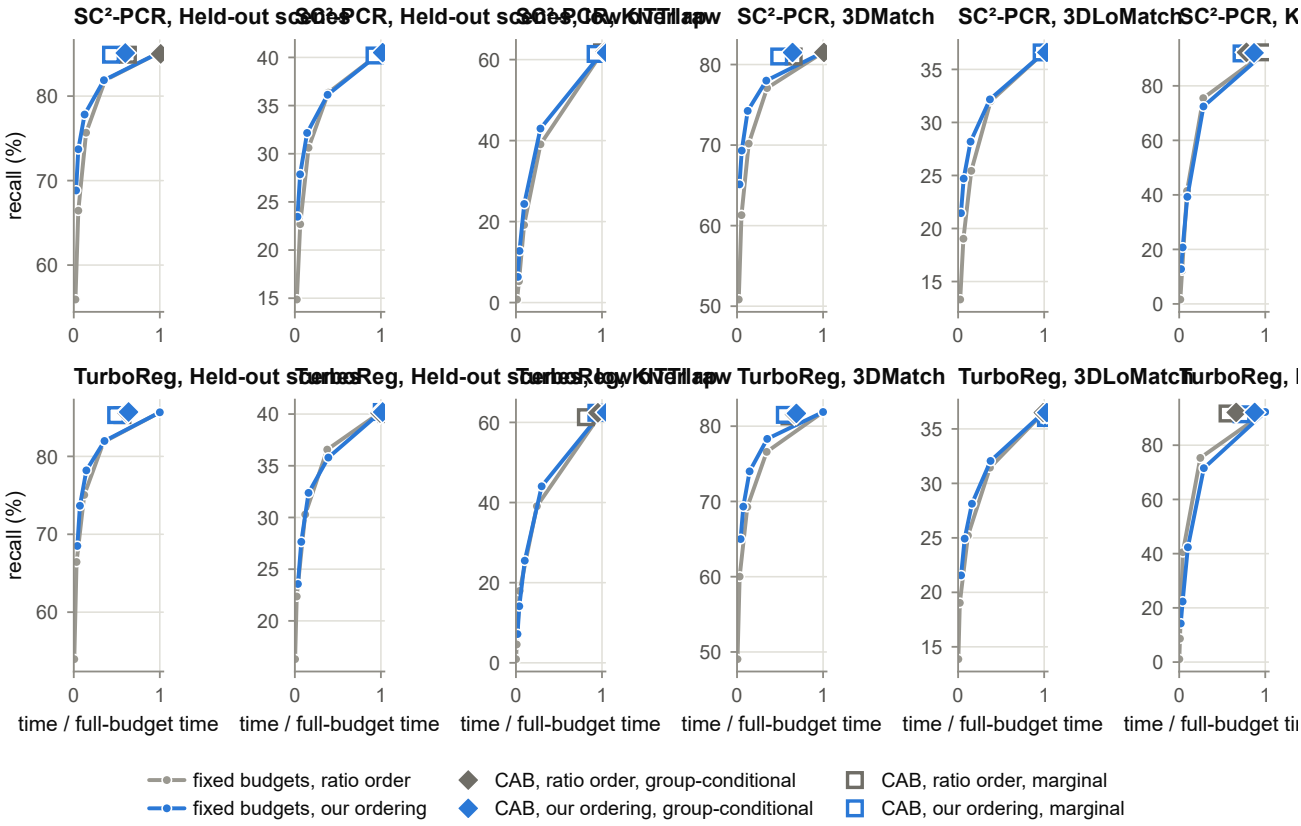


Figure 1. **Recall vs. time** on the FPFH test sets. Lines: fixed budgets $N = 250 \ldots 4000$ under ratio order (grey) and our compatibility order (blue), time relative to the full budget of the same run. Markers: certified CAB operating points, coloured by ordering (filled diamond: group-conditional certificate; open square: marginal certificate).

How many calibration pairs a certificate needs. With no loss observed among n calibration pairs, the exact binomial test certifies a 1 % level at $\delta = 0.1$ only if $n \geq 230$, and the group-conditional certificate needs 339 pairs in each of its three groups at level $\delta/3$. Table 15 checks this on random subsets of the calibration pools of 3DMatch and KITTI (20 subsets per size, from 100 pairs to the whole pool), re-certifying the thresholds of CAB's selected configuration and evaluating each certificate once on test. The group-conditional certificate first appears between 900 and 1,200 pairs, as the bound predicts ($3 \times 339 = 1{,}017$); below that, CAB runs the full budget. The marginal certificate appears from 300 pairs, but on KITTI up to 51 % of the marginal certificates (TurboReg) exceed the 1 % level on test, while no group-conditional certificate does at any size (largest risk 0.54 %). The safety of the group-conditional certificate is paid for with a calibration set of about a thousand pairs.

Checks of the guarantee. Proposition 1 bounds the risk of every difficulty group, and hence of any mixture of easy and hard pairs; the worst mixture puts all its weight on the group with the largest risk. Table 12 reports this worst-case risk on test. For CAB it is at most 0.67 % on the FPFH test sets and 2.10 % overall; the marginal certificate, evaluated on the same groups, reaches up to 3.24 %, which a test set made only of such pairs would lose. Of the 60 test groups, 2 exceed 1 % for CAB (at most 2.10 %, 9 of 428 pairs); the certificate bounds the expected risk of each group, and in a group of 428 pairs a single pair moves the realized risk by 0.2 pp. Scenes are not certified groups, and with few pairs each their risks scatter more (up to the value in the table). Finally, re-fitting the controller on ten bootstrap resamples of the fitting pairs and re-certifying its thresholds changes the speed-up moderately (development benchmarks); 1 of the 120 re-fitted certificates exceed 1 % on test (at most 1.08 %), consistent with a guarantee that holds with probability $1 - \delta = 0.9$.

Sensitivity to the risk level and the number of groups. Table 14 repeats the selection with other risk levels ε and numbers of groups G (compatibility order; for each setting the candidate is chosen on the fitting pairs, certified on the calibration pool and evaluated once on test). The speed-up grows with ε, from 1.55× at 1 % to 2.25× at 5 % for SC^2-PCR on 3DMatch, and the realized risk stays below ε at every

level; at $\varepsilon = 0.5\,\%$ the pools cannot certify anything and CAB runs the full budget. With one or two groups the chosen candidates also stay within the level on these test sets (at most 0.90 %) and are faster, but only the group-conditional certificate covers other mixes of difficulty (Tab. 12); four or more groups need more calibration pairs than the pools contain, and CAB runs the full budget.

Bounding the worst case. The pre-registered worst-case variant admits a ladder only if running all its stages costs at most $1 + \rho$ times the full budget on 95 % of the fitting pairs, and otherwise leaves the selection unchanged (Tab. 13). With $\rho = 0.25$, the outdoor ladder changes to the $\times 4$ ladder, which removes CAB's slowdown on the KITTI raw drives (1.00× and 1.01× instead of 0.95× and 0.95×, offline) at a smaller speed-up on KITTI odometry, while the indoor selection is unchanged; with $\rho = 0.10$ the indoor speed-ups shrink or vanish. Every variant stays within the risk level on every test set.

F. Equal-time comparison: details

Variants. *Fewer seeds*: SC^2-PCR on all correspondences with a maximum seed ratio of 0.025 or 0.01 instead of 0.2 (non-maximum suppression already limits the seeds to a few hundred). *Fewer cliques*: TurboReg with $K_1 = 500$ or 250 instead of 2,000. *Uniform sampling*: a random subset of N correspondences. *FastMAC-style sampling* [11]: compatibility weights $W_{ij} = 1 - S_{ij}^2/(2d^2)$, kept if above 0.999, on all correspondences (second-order graph $W \odot WW$ as published, or the first-order graph W as a cheaper variant); the degree signal $s = W\mathbf{1}$ is high-pass filtered, $f = (\mathrm{Diag}(s) - W)s$, and N correspondences are drawn without replacement with probability $\propto f_i^2$. The signal is computed once per pair and charged to every budget. Recall is measured on every second test pair (by index; 1,046 pairs of the held-out scenes and 891 3DLoMatch pairs; FastMAC-style second order on every fourth, fewer seeds/cliques on every eighth), and time live on 300 test pairs.

G. Certified adaptive pose search (CAPS)

The certification machinery is not specific to correspondence budgets. We also applied it to a pose search, which replaces a robust estimator altogether, and compare it with RANSAC.

Search. Rotations come from a scrambled Sobol sequence [9] mapped to SO(3) [8], in nested grids of 256 to 2,048 rotations. For a rotation R, every correspondence votes for the translation $\mathbf{q}_i - R\mathbf{p}_i$ (source centred); votes are hashed into cells whose side grows with the grid's covering angle, and the densest of two staggered cell grids scores R. The best basins after pose-space non-maximum suppression are refined by annealed least squares. The anytime schedule (256 rotations, 1 basin) → (256, 4) → (1,024, 4) → (2,048, 16) was chosen on validation; its reference is the schedule run to completion. *Progressive voting* scores rotations on growing prefixes of the ratio-sorted correspondences and keeps the best quarter in each round, successive halving [5] applied to rotation search (CAPS-fast); full voting scores all of them (CAPS-accurate). The controller is the Markov controller of Sec. 3.3 of the main paper, certified with the marginal certificate; cost is counted in correspondence evaluations (CE).

Results. On the same 1,000 mutual FPFH correspondences and the official RMSE criterion, CAPS-fast is 1.29× faster than RANSAC with its classical stopping rule on 3DMatch at equal recall and 1.21× faster on 3DLoMatch, where it registers +2.8 pp more pairs (Tab. 17); CAPS-accurate registers +4.8 pp more 3DLoMatch pairs than RANSAC, but runs at 0.30–0.40× its speed. The Markov controller is the cheapest certified family and the only one non-inferior on 3DMatch (Tab. 18). RANSAC's classical rule spends 22 % and 58 % of its budget on pairs with fewer than ten true correspondences, which it almost never registers (Tab. 19). Against fixed search, CAPS is cheaper than the largest configuration run once on 3DMatch (1.66–2.14× fewer CE) but not on 3DLoMatch (0.84–0.93×).

H. Evaluation pitfalls

Adaptive methods are judged by modest cost ratios, so the evaluation can create or erase the result. Table 20 lists the pitfalls we encountered in this study, with the rule we now follow; three of them changed our own numbers.

I. Reproducibility

Every number, table and figure of the paper and of this supplementary materialis generated from per-pair logs by one script, so no value is typed by hand. Protocols v2 and v3 are pre-registered in `PROTOCOL.md`; `scripts/run_v3.sh` reproduces the fresh blind test, and `scripts/validate_numbers.py` recomputes every headline number from the per-pair logs with its own cost accounting. The code covers data preparation (including the selective KITTI downloads, the KITTI raw pairs and their ICP ground truth, and the FCGF re-implementation with its verification against the released features), log construction for every split and variant, selection on the fitting pairs and certification on the calibration pairs, single-look test evaluation, the live benchmarks, and the generation of the paper's assets.

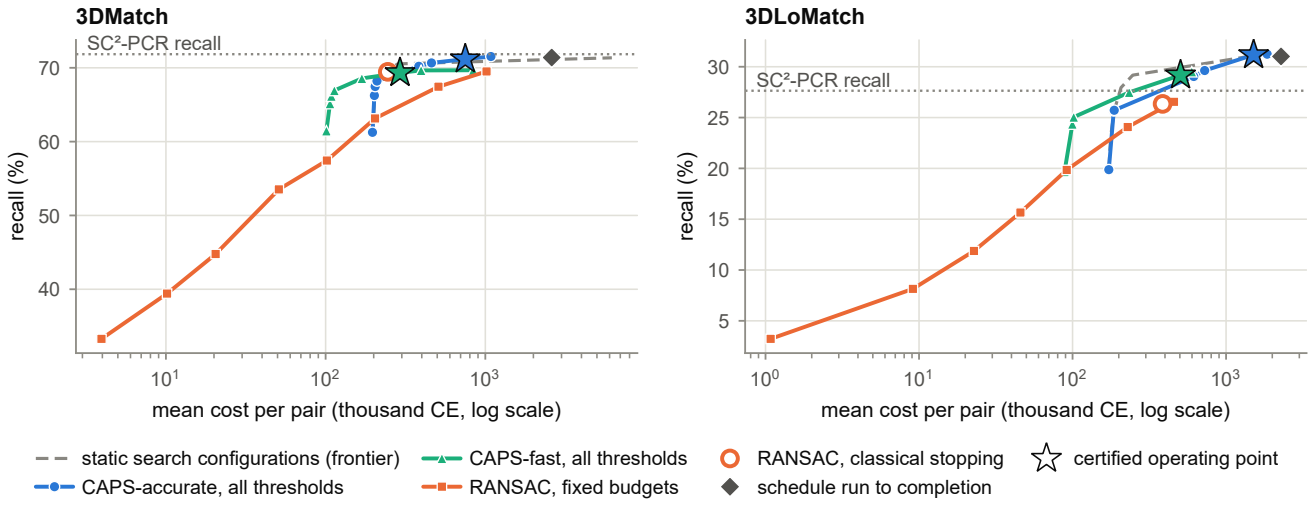


Figure 2. **CAPS on the recall–cost plane.** Lines trace all thresholds of each controller; stars mark the certified operating points.

Table 3. **All certified configurations of CAB without the gate** (protocol v2) on the test sets. Ord.: ratio or compatibility order; Cert.: marginal or group-conditional. Loss: recall lost against the full budget (pp) with its one-sided 95 % upper bound (✓: non-inferior, margin 1 pp, 2 pp at low overlap). Speed-ups: offline (full test set) and live (300 test pairs per set, group-conditional only). Fr.: recall above the best mixture of fixed budgets of the same ordering at equal time (pp).

Data	Estimator	Ord.	Cert.	Ladder	RR (%)	Loss [UB]	Off.	Live	Risk (%)	Fr.
Held-out scenes	SC^2-PCR	ratio	marginal	x4	84.9	0.24 [0.43] ✓	1.59×	–	0.29	+1.7
Held-out scenes	SC^2-PCR	ratio	group	250-500-4000	85.0	0.14 [0.33] ✓	0.99×	0.95×	0.19	−0.1
Held-out scenes	SC^2-PCR	compat.	marginal	500-4000	84.9	0.29 [0.53] ✓	2.31×	–	0.33	+2.6
Held-out scenes	SC^2-PCR	compat.	group	500-4000	85.1	0.05 [0.19] ✓	1.67×	1.56×	0.10	+2.0
Held-out scenes	TurboReg	ratio	marginal	x4	85.3	0.33 [0.57] ✓	1.77×	–	0.38	+2.2
Held-out scenes	TurboReg	ratio	group	250-500-4000	85.6	0.10 [0.29] ✓	1.62×	1.64×	0.19	+2.2
Held-out scenes	TurboReg	compat.	marginal	250-500-4000	85.3	0.33 [0.57] ✓	2.03×	–	0.38	+2.6
Held-out scenes	TurboReg	compat.	group	500-4000	85.7	0.00 [0.10] ✓	1.57×	1.49×	0.05	+2.1
Held-out scenes, low overlap	SC^2-PCR	ratio	marginal	x4	40.4	0.11 [0.33] ✓	0.88×	–	0.11	−0.1
Held-out scenes, low overlap	SC^2-PCR	ratio	group	250-500-4000	40.3	0.22 [0.45] ✓	0.94×	0.89×	0.22	−0.2
Held-out scenes, low overlap	SC^2-PCR	compat.	marginal	500-4000	40.2	0.33 [0.66] ✓	1.08×	–	0.33	+0.2
Held-out scenes, low overlap	SC^2-PCR	compat.	group	500-4000	40.5	0.00 [0.00] ✓	0.98×	0.94×	0.00	+0.0
Held-out scenes, low overlap	TurboReg	ratio	marginal	x4	40.0	0.22 [0.55] ✓	0.94×	–	0.22	−0.2
Held-out scenes, low overlap	TurboReg	ratio	group	250-500-4000	40.1	0.11 [0.33] ✓	1.01×	1.01×	0.11	−0.0
Held-out scenes, low overlap	TurboReg	compat.	marginal	250-500-4000	40.2	0.00 [0.22] ✓	0.99×	–	0.11	+0.0
Held-out scenes, low overlap	TurboReg	compat.	group	500-4000	40.2	0.00 [0.00] ✓	0.98×	0.95×	0.00	+0.0
KITTI raw	SC^2-PCR	ratio	marginal	full	61.8	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
KITTI raw	SC^2-PCR	ratio	group	250-2000-4000	61.8	−0.08 [0.00] ✓	0.97×	0.89×	0.00	+0.1
KITTI raw	SC^2-PCR	compat.	marginal	250-1000-2000-4000	61.4	0.38 [0.69] ✓	1.08×	–	0.46	+1.6
KITTI raw	SC^2-PCR	compat.	group	250-2000-4000	61.7	0.08 [0.23] ✓	0.95×	0.93×	0.08	−0.1
KITTI raw	TurboReg	ratio	marginal	250-1000-2000-4000	61.3	1.30 [1.83] ✗	1.23×	–	1.37	+4.4
KITTI raw	TurboReg	ratio	group	250-1000-2000-4000	62.4	0.15 [0.38] ✓	1.05×	1.00×	0.15	+1.3
KITTI raw	TurboReg	compat.	marginal	500-2000-4000	62.4	0.23 [0.61] ✓	1.07×	–	0.46	+1.6
KITTI raw	TurboReg	compat.	group	250-2000-4000	62.5	0.08 [0.38] ✓	0.95×	0.91×	0.31	−0.1
KITTI raw (FCGF)	SC^2-PCR	ratio	marginal	full	92.6	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
KITTI raw (FCGF)	SC^2-PCR	ratio	group	full	92.6	0.00 [0.00] ✓	1.00×	1.00×	0.00	+0.0
KITTI raw (FCGF)	SC^2-PCR	compat.	marginal	full	92.6	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
KITTI raw (FCGF)	SC^2-PCR	compat.	group	full	92.6	0.00 [0.00] ✓	1.00×	1.00×	0.00	+0.0
KITTI raw (FCGF)	TurboReg	ratio	marginal	full	92.1	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
KITTI raw (FCGF)	TurboReg	ratio	group	full	92.1	0.00 [0.00] ✓	1.00×	1.00×	0.00	+0.0
KITTI raw (FCGF)	TurboReg	compat.	marginal	full	92.1	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
KITTI raw (FCGF)	TurboReg	compat.	group	full	92.1	0.00 [0.00] ✓	1.00×	1.00×	0.00	+0.0
3DMatch	SC^2-PCR	ratio	marginal	x4	81.0	0.49 [0.80] ✓	1.52×	–	0.49	+1.8
3DMatch	SC^2-PCR	ratio	group	250-500-4000	81.5	0.06 [0.18] ✓	0.99×	0.93×	0.06	−0.1
3DMatch	SC^2-PCR	compat.	marginal	500-4000	81.0	0.49 [0.86] ✓	2.04×	–	0.68	+2.2
3DMatch	SC^2-PCR	compat.	group	500-4000	81.5	0.00 [0.18] ✓	1.55×	1.58×	0.12	+1.9
3DMatch	TurboReg	ratio	marginal	x4	81.3	0.62 [0.99] ✓	1.64×	–	0.62	+2.5
3DMatch	TurboReg	ratio	group	250-500-4000	81.6	0.31 [0.62] ✓	1.50×	1.59×	0.37	+2.4
3DMatch	TurboReg	compat.	marginal	250-500-4000	81.5	0.43 [0.68] ✓	1.80×	–	0.43	+2.0
3DMatch	TurboReg	compat.	group	500-4000	81.7	0.18 [0.37] ✓	1.45×	1.42×	0.18	+1.5
3DLoMatch	SC^2-PCR	ratio	marginal	x4	36.6	0.00 [0.00] ✓	0.88×	–	0.00	+0.0
3DLoMatch	SC^2-PCR	ratio	group	250-500-4000	36.6	0.06 [0.17] ✓	0.94×	0.90×	0.06	−0.1
3DLoMatch	SC^2-PCR	compat.	marginal	500-4000	36.6	0.06 [0.17] ✓	1.03×	–	0.06	+0.2
3DLoMatch	SC^2-PCR	compat.	group	500-4000	36.6	0.06 [0.17] ✓	0.97×	0.96×	0.06	−0.1
3DLoMatch	TurboReg	ratio	marginal	x4	36.4	0.11 [0.28] ✓	0.94×	–	0.11	−0.1
3DLoMatch	TurboReg	ratio	group	250-500-4000	36.5	0.06 [0.22] ✓	1.01×	1.03×	0.11	+0.0
3DLoMatch	TurboReg	compat.	marginal	250-500-4000	36.0	0.56 [0.84] ✓	0.98×	–	0.56	−0.6
3DLoMatch	TurboReg	compat.	group	500-4000	36.5	0.06 [0.17] ✓	0.97×	0.97×	0.06	−0.1
KITTI	SC^2-PCR	ratio	marginal	full	92.3	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
KITTI	SC^2-PCR	ratio	group	250-2000-4000	92.3	0.00 [0.00] ✓	1.28×	1.18×	0.00	+5.1
KITTI	SC^2-PCR	compat.	marginal	250-1000-2000-4000	91.9	0.36 [0.90] ✓	1.38×	–	0.54	+7.2
KITTI	SC^2-PCR	compat.	group	250-2000-4000	92.1	0.18 [0.54] ✓	1.15×	1.08×	0.18	+3.5
KITTI	TurboReg	ratio	marginal	250-1000-2000-4000	91.7	0.54 [1.08] ✗	1.78×	–	0.54	+9.3
KITTI	TurboReg	ratio	group	250-1000-2000-4000	92.1	0.18 [0.54] ✓	1.51×	1.39×	0.18	+7.4
KITTI	TurboReg	compat.	marginal	500-2000-4000	91.4	0.90 [1.62] ✗	1.38×	–	0.90	+7.0
KITTI	TurboReg	compat.	group	250-2000-4000	92.1	0.18 [0.72] ✓	1.14×	1.07×	0.36	+3.4
3DMatch (FCGF)	SC^2-PCR	ratio	marginal	full	92.5	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
3DMatch (FCGF)	SC^2-PCR	ratio	group	500-4000	92.5	0.06 [0.37] ✓	1.03×	1.00×	0.31	−0.0
3DMatch (FCGF)	SC^2-PCR	compat.	marginal	250-4000	91.7	0.86 [1.42] ✗	5.59×	–	1.29	+1.2
3DMatch (FCGF)	SC^2-PCR	compat.	group	500-4000	92.2	0.37 [0.68] ✓	1.06×	1.04×	0.43	−0.3
3DMatch (FCGF)	TurboReg	ratio	marginal	500-4000	91.8	0.80 [1.29] ✗	4.69×	–	1.05	+0.3
3DMatch (FCGF)	TurboReg	ratio	group	500-4000	92.6	0.00 [0.12] ✓	0.99×	0.95×	0.06	+0.0
3DMatch (FCGF)	TurboReg	compat.	marginal	250-4000	91.9	0.74 [1.23] ✗	5.51×	–	1.05	+1.5
3DMatch (FCGF)	TurboReg	compat.	group	500-4000	92.1	0.49 [0.86] ✓	1.05×	1.01×	0.55	−0.4
3DLoMatch (FCGF)	SC^2-PCR	ratio	marginal	full	58.0	0.00 [0.00] ✓	1.00×	–	0.00	+0.0
3DLoMatch (FCGF)	SC^2-PCR	ratio	group	500-4000	57.7	0.34 [0.62] ✓	1.07×	1.01×	0.45	−0.2

Table 4. **Fixed budgets** (single runs): recall (%) at $N = 250/500/1000/2000$ under ratio and compatibility order, and the full budget.

		3DMatch					3DLoMatch					KITTI				
Estimator	Order	250	500	1k	2k	full	250	500	1k	2k	full	250	500	1k	2k	full
SC^2-PCR	ratio	50.8	61.3	70.2	77.1	81.5	13.3	19.0	25.4	32.0	36.6	1.6	12.3	41.4	75.5	92.3
SC^2-PCR	compat.	65.1	69.3	74.2	78.0	81.5	21.4	24.7	28.2	32.2	36.6	12.8	20.7	39.3	72.4	92.3
TurboReg	ratio	49.0	60.0	69.3	76.6	81.9	13.9	19.0	25.2	31.4	36.6	1.1	8.6	40.5	75.3	92.3
TurboReg	compat.	65.0	69.3	74.0	78.3	81.9	21.6	24.9	28.1	32.1	36.6	14.2	22.3	42.3	71.5	92.3

Table 5. **The final method on every test set** (protocol v3; group-conditional certificates). Clf.: controller classifier chosen on the fitting pairs; Gate: gate classifier (–: no gate, the full budget). Loss: recall lost against the full budget (pp) with its one-sided 95 % upper bound (✓: non-inferior). Off.: offline speed-up (full test set); Live: live speed-up (300 pairs per set; †full budget by construction, not timed). Gated: share of pairs sent to the full budget (%). Fresh scenes are the blind test of protocol v3; the other sets are not blind for this method.

Test set	Estimator	Ord.	Ladder	Clf.	Gate	RR (%)	Loss [UB]	Off.	Live	Ops	Risk (%)	Gated
Fresh scenes	SC^2-PCR	ratio	500-4000	quad.	trees	93.0	0.05 [0.15] ✓	1.85×	1.62×	1.93×	0.05	30
		compat.	500-4000	log. 0.1	trees	93.1	0.00 [0.00] ✓	1.90×	1.68×	1.96×	0.00	26
	TurboReg	ratio	x4	log. 0.1	trees	92.9	0.05 [0.15] ✓	1.66×	1.57×	1.75×	0.05	28
		compat.	500-4000	quad.	trees	92.9	0.00 [0.00] ✓	2.19×	2.10×	2.45×	0.00	20
Fresh scenes, low overlap	SC^2-PCR	ratio	500-4000	quad.	trees	52.9	0.00 [0.00] ✓	1.05×	1.01×	1.05×	0.00	80
		compat.	500-4000	log. 0.1	trees	52.9	0.00 [0.00] ✓	1.04×	1.01×	1.05×	0.00	75
	TurboReg	ratio	x4	log. 0.1	trees	51.8	0.00 [0.00] ✓	1.01×	1.00×	1.04×	0.00	75
		compat.	500-4000	quad.	trees	51.6	0.15 [0.46] ✓	1.08×	1.05×	1.10×	0.15	66
Held-out scenes	SC^2-PCR	ratio	500-4000	quad.	trees	85.0	0.14 [0.33] ✓	1.68×	1.41×	1.66×	0.19	35
		compat.	500-4000	trees	trees	85.0	0.19 [0.38] ✓	1.82×	1.59×	1.83×	0.24	23
	TurboReg	ratio	x4	quad.	trees	85.7	0.00 [0.10] ✓	1.55×	1.52×	1.63×	0.05	30
		compat.	500-4000	trees	log.	85.3	0.38 [0.62] ✓	1.90×	1.87×	2.09×	0.38	19
Held-out scenes, low overlap	SC^2-PCR	ratio	500-4000	quad.	trees	40.5	0.00 [0.00] ✓	1.02×	0.97×	1.02×	0.00	80
		compat.	500-4000	trees	trees	40.4	0.11 [0.33] ✓	1.03×	0.96×	1.04×	0.11	58
	TurboReg	ratio	x4	quad.	trees	40.2	0.00 [0.00] ✓	1.00×	0.99×	1.03×	0.00	76
		compat.	500-4000	trees	log.	40.2	0.00 [0.00] ✓	1.05×	1.01×	1.08×	0.00	46
KITTI raw	SC^2-PCR	ratio	500-2000-4000	log. 0.1	log.	61.8	−0.08 [0.00] ✓	1.04×	0.99×	1.13×	0.00	58
		compat.	500-2000-4000	quad.	log.	61.6	0.15 [0.31] ✓	1.10×	1.08×	1.18×	0.15	61
	TurboReg	ratio	250-1000-2000-4000	log.	log.	62.4	0.15 [0.38] ✓	1.15×	1.10×	1.24×	0.15	58
		compat.	250-2000-4000	log. 0.1	log.	62.7	−0.08 [0.00] ✓	1.06×	1.03×	1.13×	0.00	71
KITTI raw (FCGF)	SC^2-PCR	ratio	full	log. 0.1	–	92.6	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0
		compat.	full	log.	–	92.6	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0
	TurboReg	ratio	full	log. 0.1	–	92.1	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0
		compat.	full	log.	–	92.1	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0
3DMatch	SC^2-PCR	ratio	500-4000	quad.	trees	81.3	0.25 [0.49] ✓	1.59×	1.50×	1.61×	0.25	38
		compat.	500-4000	trees	trees	81.1	0.37 [0.62] ✓	1.71×	1.55×	1.75×	0.37	24
	TurboReg	ratio	x4	quad.	trees	81.6	0.25 [0.49] ✓	1.48×	1.50×	1.56×	0.25	32
		compat.	500-4000	trees	log.	81.3	0.55 [0.86] ✓	1.77×	1.75×	1.94×	0.55	17
3DLoMatch	SC^2-PCR	ratio	500-4000	quad.	trees	36.5	0.11 [0.22] ✓	1.02×	1.01×	1.02×	0.11	82
		compat.	500-4000	trees	trees	36.6	0.06 [0.22] ✓	1.03×	0.99×	1.03×	0.11	55
	TurboReg	ratio	x4	quad.	trees	36.5	0.06 [0.17] ✓	1.01×	1.01×	1.03×	0.06	79
		compat.	500-4000	trees	log.	36.5	0.06 [0.28] ✓	1.04×	1.03×	1.08×	0.17	43
KITTI	SC^2-PCR	ratio	500-2000-4000	log. 0.1	log.	92.3	0.00 [0.00] ✓	1.20×	1.13×	1.39×	0.00	35
		compat.	500-2000-4000	quad.	log.	91.9	0.36 [0.90] ✓	1.30×	1.24×	1.54×	0.36	26
	TurboReg	ratio	250-1000-2000-4000	log.	log.	92.1	0.18 [0.54] ✓	1.51×	1.40×	1.73×	0.18	25
		compat.	250-2000-4000	log. 0.1	log.	92.1	0.18 [0.54] ✓	1.18×	1.12×	1.35×	0.18	40
3DMatch (FCGF)	SC^2-PCR	ratio	500-4000	log.	trees	92.5	0.06 [0.37] ✓	1.05×	1.03×	1.04×	0.31	73
		compat.	500-4000	quad.	trees	92.4	0.18 [0.37] ✓	1.05×	1.04×	1.04×	0.18	81
	TurboReg	ratio	500-4000	log. 0.1	trees	92.5	0.06 [0.25] ✓	1.02×	1.00×	1.02×	0.12	89
		compat.	500-4000	trees	trees	92.2	0.37 [0.68] ✓	1.07×	1.07×	1.09×	0.55	74
3DLoMatch (FCGF)	SC^2-PCR	ratio	500-4000	log.	trees	57.7	0.28 [0.56] ✓	1.09×	1.01×	1.08×	0.39	50
		compat.	500-4000	quad.	trees	58.0	0.00 [0.17] ✓	1.08×	1.01×	1.07×	0.11	65
	TurboReg	ratio	500-4000	log. 0.1	trees	58.4	0.00 [0.00] ✓	1.03×	1.02×	1.04×	0.00	81
		compat.	500-4000	trees	trees	58.3	0.11 [0.39] ✓	1.12×	1.05×	1.17×	0.28	43
KITTI (FCGF)	SC^2-PCR	ratio	full	log. 0.1	–	93.5	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0
		compat.	full	log.	–	93.5	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0
	TurboReg	ratio	full	log. 0.1	–	93.5	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0
		compat.	full	log.	–	93.5	0.00 [0.00] ✓	1.00×	1.00†	1.00×	0.00	0

Table 6. **Protocol v2: CAB without the gate** (logistic controller) on its blind test sets, evaluated once (group-conditional certificates; columns as in Table 1 of the main paper). With FCGF on the KITTI raw drives, CAB runs the full budget.

	Held-out scenes (2,092 pairs)					Held-out scenes, low overlap (908)					KITTI raw drives (1,310)				
Method	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk
FPFH correspondences															
SC²-PCR															
Full budget (standard)	85.2	–	1.00×	1.00×	–	40.5	–	1.00×	1.00×	–	61.8	–	1.00×	1.00×	–
CAB, ratio order	85.0	0.14 [0.33] ✓	0.95×	1.05×	0.19	40.3	0.22 [0.45] ✓	0.89×	1.01×	0.22	61.8	−0.08 [0.00] ✓	0.89×	1.18×	0.00
CAB (ours)	**85.1**	0.05 [0.19] ✓	**1.56**×	**1.69**×	0.10	**40.5**	0.00 [0.00] ✓	**0.94**×	**1.02**×	0.00	**61.7**	0.08 [0.23] ✓	**0.93**×	**1.14**×	0.08
TurboReg															
Full budget (standard)	85.7	–	1.00×	1.00×	–	40.2	–	1.00×	1.00×	–	62.6	–	1.00×	1.00×	–
CAB, ratio order	85.6	0.10 [0.29] ✓	1.64×	1.64×	0.19	40.1	0.11 [0.33] ✓	1.01×	1.03×	0.11	62.4	0.15 [0.38] ✓	1.00×	1.21×	0.15
CAB (ours)	**85.7**	0.00 [0.10] ✓	**1.49**×	**1.72**×	0.05	**40.2**	0.00 [0.00] ✓	**0.95**×	**1.05**×	0.00	**62.5**	0.08 [0.38] ✓	**0.91**×	**1.15**×	0.31

Table 7. **Protocol v2: CAB without the gate** on the development benchmarks (columns as in the previous table). †CAB runs the full budget.

	3DMatch (1,623 pairs)					3DLoMatch (1,781)					KITTI odometry (555)				
Method	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk
FPFH correspondences															
SC²-PCR															
Full budget (standard)	81.5	–	1.00×	1.00×	–	36.6	–	1.00×	1.00×	–	92.3	–	1.00×	1.00×	–
CAB, ratio order	81.5	0.06 [0.18] ✓	0.93×	1.05×	0.06	36.6	0.06 [0.17] ✓	0.90×	1.00×	0.06	92.3	0.00 [0.00] ✓	1.18×	1.63×	0.00
CAB (ours)	**81.5**	0.00 [0.18] ✓	**1.58**×	**1.61**×	0.12	**36.6**	0.06 [0.17] ✓	**0.96**×	**1.01**×	0.06	**92.1**	0.18 [0.54] ✓	**1.08**×	**1.42**×	0.18
TurboReg															
Full budget (standard)	81.9	–	1.00×	1.00×	–	36.6	–	1.00×	1.00×	–	92.3	–	1.00×	1.00×	–
CAB, ratio order	81.6	0.31 [0.62] ✓	1.59×	1.52×	0.37	36.5	0.06 [0.22] ✓	1.03×	1.03×	0.11	92.1	0.18 [0.54] ✓	1.39×	1.78×	0.18
CAB (ours)	**81.7**	0.18 [0.37] ✓	**1.42**×	**1.58**×	0.18	**36.5**	0.06 [0.17] ✓	**0.97**×	**1.04**×	0.06	**92.1**	0.18 [0.72] ✓	**1.07**×	**1.41**×	0.36
FCGF correspondences															
SC²-PCR															
Full budget (standard)	92.5	–	1.00×	1.00×	–	58.0	–	1.00×	1.00×	–	93.5	–	1.00×	1.00×	–
CAB, ratio order	92.5	0.06 [0.37] ✓	1.00×	1.06×	0.31	57.7	0.34 [0.62] ✓	1.01×	1.10×	0.45	93.5	0.00 [0.00] ✓	1.00×†	1.00×	0.00
CAB (ours)	**92.2**	0.37 [0.68] ✓	**1.04**×	**1.07**×	0.43	**57.7**	0.28 [0.56] ✓	**1.08**×	**1.15**×	0.45	**93.5**	0.00 [0.00] ✓	**1.00**×†	**1.00**×	0.00
TurboReg															
Full budget (standard)	92.6	–	1.00×	1.00×	–	58.4	–	1.00×	1.00×	–	93.5	–	1.00×	1.00×	–
CAB, ratio order	92.6	0.00 [0.12] ✓	0.95×	1.03×	0.06	58.3	0.11 [0.28] ✓	0.99×	1.06×	0.11	93.5	0.00 [0.00] ✓	1.00×†	1.00×	0.00
CAB (ours)	**92.1**	0.49 [0.86] ✓	**1.01**×	**1.10**×	0.55	**58.2**	0.17 [0.51] ✓	**1.03**×	**1.19**×	0.39	**93.5**	0.00 [0.00] ✓	**1.00**×†	**1.00**×	0.00

Table 8. **Success at $N = 250$ by true anchors** (SC^2-PCR, %). Full: full-budget success.

Data	True anchors	Pairs	Full	Ratio	Compat.	Gain
3DM+3DLoM	0	908	10.9	0.0	**0.2**	+0.2
	1–4	697	36.7	1.1	**2.0**	+0.9
	5–19	717	76.6	15.8	**49.4**	+33.6
	20–49	552	98.0	74.6	**97.6**	+23.0
	$\geq$50	530	100.0	99.8	**100.0**	+0.2
KITTI	0	281	87.9	0.0	**2.5**	+2.5
	1–4	189	96.3	0.5	**10.1**	+9.5
	5–19	75	97.3	2.7	**48.0**	+45.3
	20–49	10	100.0	60.0	**90.0**	+30.0

Table 9. **Per-pair latency** (live, ms; full budget / CAB without the gate, compatibility order): 50th, 90th and 99th percentiles, and the share of pairs on which CAB is slower than the full budget (%).

	Pairs	p50	p90	p99	Slower
SC^2-PCR, Held-out scenes	300	717 / 308	801 / 862	879 / 933	49
TurboReg, Held-out scenes	300	1,114 / 886	1,132 / 1,229	1,179 / 1,293	53
SC^2-PCR, Held-out scenes, low overlap	300	652 / 684	786 / 854	837 / 913	94
TurboReg, Held-out scenes, low overlap	300	978 / 1,037	1,127 / 1,229	1,204 / 1,307	93
SC^2-PCR, 3DMatch	300	727 / 251	781 / 843	842 / 929	50
TurboReg, 3DMatch	300	1,115 / 903	1,133 / 1,230	1,183 / 1,295	57
SC^2-PCR, 3DLoMatch	300	722 / 778	792 / 858	944 / 980	89
TurboReg, 3DLoMatch	300	1,110 / 1,190	1,124 / 1,229	1,180 / 1,283	90
SC^2-PCR, KITTI raw	300	1,079 / 1,436	1,120 / 1,512	1,195 / 1,562	73
TurboReg, KITTI raw	300	1,108 / 1,487	1,143 / 1,558	1,211 / 1,643	74
SC^2-PCR, KITTI	300	1,085 / 1,423	1,132 / 1,517	1,181 / 1,597	58
TurboReg, KITTI	300	1,108 / 1,418	1,145 / 1,553	1,226 / 1,644	58
SC^2-PCR, 3DMatch (FCGF)	300	687 / 728	732 / 775	869 / 846	83
TurboReg, 3DMatch (FCGF)	300	1,135 / 1,238	1,179 / 1,284	1,242 / 1,341	86
SC^2-PCR, 3DLoMatch (FCGF)	300	681 / 709	730 / 774	890 / 868	79
TurboReg, 3DLoMatch (FCGF)	300	1,117 / 1,217	1,142 / 1,253	1,195 / 1,306	83

Table 10. **Headroom.** Speed-up over the full budget of CAB (group-conditional, offline), of two oracles on the same ladder: one stops at the first correct stage but cannot detect failures (Acc.), the other also stops unsolvable pairs after the first stage (Full); share of unsolvable pairs and of their full-budget time (%).

	Ratio order			Compat. order			Unsolvable	
	CAB	Acc.	Full	CAB	Acc.	Full	pairs	time
SC^2-PCR, Held-out scenes	0.99	2.50	4.33	1.67	2.92	5.52	14	16
TurboReg, Held-out scenes	1.62	2.71	4.69	1.57	2.81	4.96	14	15
SC^2-PCR, Held-out scenes, low overlap	0.94	1.15	4.40	0.98	1.23	4.73	58	60
TurboReg, Held-out scenes, low overlap	1.01	1.25	5.06	0.98	1.22	4.53	58	59
SC^2-PCR, KITTI raw	0.97	1.09	2.42	0.95	1.18	2.59	38	39
TurboReg, KITTI raw	1.05	1.18	2.69	0.95	1.19	2.55	37	38
SC^2-PCR, 3DMatch	0.99	2.21	3.90	1.55	2.64	4.99	18	19
TurboReg, 3DMatch	1.50	2.34	4.05	1.45	2.54	4.63	17	18
SC^2-PCR, 3DLoMatch	0.94	1.10	4.43	0.97	1.18	4.93	62	65
TurboReg, 3DLoMatch	1.01	1.17	5.10	0.97	1.16	4.71	62	65
SC^2-PCR, KITTI	1.28	1.88	2.33	1.15	1.86	2.26	8	8
TurboReg, KITTI	1.51	2.41	3.08	1.14	1.83	2.21	7	7
SC^2-PCR, Fresh scenes	1.77	2.66	3.38	1.85	3.59	4.89	7	8
TurboReg, Fresh scenes	1.69	3.83	5.78	1.61	3.22	4.37	7	8
SC^2-PCR, Fresh scenes, low overlap	1.00	1.19	2.79	1.00	1.31	3.44	47	48
TurboReg, Fresh scenes, low overlap	0.97	1.30	4.26	1.01	1.26	3.32	47	49

Table 11. **Ordering ablation** on the 3DMatch validation scenes: inliers among the first 250 correspondences and success (%) at $N = 250/500/1000$. Full budget: 71.4 % (SC^2-PCR) and 72.2 % (TurboReg).

		SC^2-PCR			TurboReg		
Ordering	Inl.@250	250	500	1k	250	500	1k
Lowe ratio	27.8	51.3	61.2	67.5	51.7	61.2	67.2
One pass	82.6	52.8	57.8	63.8	52.9	58.2	63.8
Two passes (ours)	104.7	58.8	62.6	66.8	59.4	62.9	66.6

Table 12. **Checks of the guarantee** (compatibility order, selected configurations; risks in %). Worst mix: largest test risk over the three difficulty groups, *i.e*. the risk of the worst mixture of easy and hard pairs, for CAB and for the marginal certificate. Scene: largest per-scene risk of CAB. Re-fit: controller re-fitted on 10 bootstrap resamples of the training pairs: mean offline speed-up [min, max] and largest risk.

	Worst mix			Re-fit	
	CAB	marg.	Scene	speed-up	risk
SC^2-PCR, Held-out scenes	0.22	0.72	0.62	–	–
SC^2-PCR, Held-out, low overlap	0.00	1.20	0.00	–	–
SC^2-PCR, 3DMatch	0.18	1.66	1.85	1.55× [1.49, 1.57]	0.12
SC^2-PCR, 3DLoMatch	0.07	0.07	0.19	0.97× [0.96, 0.98]	0.11
TurboReg, Held-out scenes	0.14	0.72	0.49	–	–
TurboReg, Held-out, low overlap	0.00	0.40	0.00	–	–
TurboReg, 3DMatch	0.56	1.12	1.85	1.26× [1.07, 1.49]	0.31
TurboReg, 3DLoMatch	0.07	0.72	2.04	0.96× [0.95, 0.97]	0.06
SC^2-PCR, KITTI raw	0.16	0.66	3.33	–	–
SC^2-PCR, KITTI	0.49	0.88	0.33	1.18× [1.11, 1.36]	1.08
TurboReg, KITTI raw	0.67	0.82	4.35	–	–
TurboReg, KITTI	0.51	1.67	0.65	1.21× [1.09, 1.36]	0.90
SC^2-PCR, 3DMatch (FCGF)	1.62	3.24	1.30	1.06× [1.02, 1.09]	0.62
SC^2-PCR, 3DLoMatch (FCGF)	0.52	2.48	1.38	1.14× [1.09, 1.20]	0.56
TurboReg, 3DMatch (FCGF)	2.10	2.80	1.30	1.06× [1.01, 1.08]	0.74
TurboReg, 3DLoMatch (FCGF)	0.46	2.46	1.38	1.14× [1.06, 1.18]	0.73
SC^2-PCR, KITTI raw (FCGF)	0.00	0.00	0.00	–	–
SC^2-PCR, KITTI (FCGF)	0.00	0.00	0.00	1.00× [1.00, 1.00]	0.00
TurboReg, KITTI raw (FCGF)	0.00	0.00	0.00	–	–
TurboReg, KITTI (FCGF)	0.00	0.00	0.00	1.00× [1.00, 1.00]	0.00

Table 13. **Worst-case variant** (pre-registered; compatibility order, group-conditional): selected ladder (indoor / outdoor), offline speed-up and, in parentheses, test risk (%) on every FPFH test set. ρ: bound on the whole-ladder cost relative to the full budget on 95 % of the fitting pairs (none: the paper's configuration).

			Blind			Development		
Estimator	ρ	Ladders	Held-out	Held-out, low	KITTI raw	3DMatch	3DLoMatch	KITTI
SC^2-PCR	none	500-4000 / 250-2000-4000	1.67× (0.10)	0.98× (0.00)	0.95× (0.08)	1.55× (0.12)	0.97× (0.06)	1.15× (0.18)
	0.25	500-4000 / x4	1.67× (0.10)	0.98× (0.00)	1.00× (0.15)	1.55× (0.12)	0.97× (0.06)	1.05× (0.00)
	0.10	250-4000 / 250-500-4000	1.41× (0.05)	0.99× (0.00)	0.99× (0.23)	1.34× (0.12)	0.98× (0.00)	1.00× (0.36)
TurboReg	none	500-4000 / 250-2000-4000	1.57× (0.05)	0.98× (0.00)	0.95× (0.31)	1.45× (0.18)	0.97× (0.06)	1.14× (0.36)
	0.25	500-4000 / x4	1.57× (0.05)	0.98× (0.00)	1.01× (0.08)	1.45× (0.18)	0.97× (0.06)	1.09× (0.72)
	0.10	full / 500-4000	1.00× (0.00)	1.00× (0.00)	1.05× (0.00)	1.00× (0.00)	1.00× (0.00)	1.07× (0.00)

Table 14. **Sensitivity to ε and G** (compatibility order): offline speed-up and, in parentheses, test risk (%; bold: above ε). The paper uses $\varepsilon = 1\,\%$, $G = 3$; $G = 1$ is the marginal certificate.

		3DMatch		KITTI	
ε (%)	G	SC^2-PCR	TurboReg	SC^2-PCR	TurboReg
0.5	3	0.97× (0.06)	0.98× (0.06)	1.00× (0.00)	1.00× (0.00)
1	3	1.55× (0.12)	1.45× (0.18)	1.15× (0.18)	1.14× (0.36)
2	3	1.86× (0.37)	1.92× (0.49)	1.37× (0.36)	1.41× (1.08)
5	3	2.25× (1.23)	2.19× (1.17)	1.48× (1.08)	1.52× (2.88)
1	1	2.04× (0.62)	1.80× (0.43)	1.41× (0.90)	1.34× (0.72)
1	2	1.58× (0.25)	1.88× (0.55)	1.41× (0.90)	1.35× (0.90)
1	4	1.29× (0.12)	1.28× (0.25)	1.00× (0.00)	1.00× (0.00)
1	5	1.19× (0.06)	1.19× (0.06)	1.00× (0.00)	1.00× (0.00)

Table 15. **Calibration size.** Certification on random subsets of the calibration pool (20 per size; compatibility order). Min. n: smallest subset size at which most subsets certify an early stop; $>1\,\%$: share of the certificates (all sizes) whose test risk exceeds 1 %; max: largest test risk (%).

	Group-conditional			Marginal		
Test set	min. n	$>1\,\%$	max	min. n	$>1\,\%$	max
SC^2-PCR, 3DMatch	1,200	0	0.25	300	1	1.11
SC^2-PCR, 3DLoMatch	1,200	0	0.06	300	0	0.11
TurboReg, 3DMatch	1,200	0	0.37	300	3	1.05
TurboReg, 3DLoMatch	1,200	0	0.06	300	0	0.17
SC^2-PCR, KITTI	1,200	0	0.18	300	0	0.54
TurboReg, KITTI	900	0	0.54	300	51	2.88

Table 16. **3DLoMatch, equal-time comparison.** Time: offline time relative to the full budget on the same pairs (sampling signal included). Δ: recall minus that of the full budget on the same pairs (pp; full budget on all 891 pairs: 37.5 % for SC^2-PCR, 37.7 % for TurboReg). (a)/(b): seed ratio 0.025/0.01 or $K_1 = 500/250$.

	SC^2-PCR		TurboReg	
Variant	time	Δ	time	Δ
CAB (ours)	1.00	+0.0	0.99	−0.1
Uniform sampling				
N=250	0.02	−24.6	0.00	−25.5
N=500	0.06	−14.8	0.01	−15.0
N=1,000	0.16	−7.2	0.10	−8.8
N=2,000	0.37	−3.7	0.35	−3.4
FastMAC-style, 1st-order graph				
N=250	0.40	−12.7	0.23	−12.0
N=500	0.44	−8.8	0.26	−8.0
N=1,000	0.53	−5.9	0.36	−6.5
N=2,000	0.75	−2.6	0.58	−2.2
FastMAC-style, 2nd-order graph				
N=250	1.03	−7.2	0.62	−7.8
N=500	1.06	−3.8	0.65	−3.8
N=1,000	1.15	−4.3	0.75	−3.8
N=2,000	1.36	−1.1	0.96	−2.2
Fixed budget, ratio order				
N=250	0.02	−23.5	0.00	−24.0
N=500	0.06	−18.6	0.02	−18.2
N=1,000	0.15	−11.8	0.11	−12.0
N=2,000	0.38	−5.5	0.37	−5.7
Fixed budget, compat. order				
N=250	0.03	−16.4	0.04	−16.4
N=500	0.07	−13.0	0.08	−13.2
N=1,000	0.15	−9.3	0.17	−10.0
N=2,000	0.38	−5.2	0.39	−5.5
Fewer seeds (SC^2-PCR) / cliques (TurboReg)				
(a)	0.84	−3.6	0.69	−1.3
(b)	0.81	−7.2	0.65	−1.3

Table 17. **CAPS against RANSAC and fixed search** (official recall, %, scene-averaged). ΔRR: CAPS minus benchmark (pp; exact McNemar p where paired outcomes were logged). CE ratio: benchmark CE / CAPS CE. Live: benchmark time / CAPS time, single process, same pairs. Values $> 1\times$ favour CAPS.

	3DMatch (1,279 pairs)				3DLoMatch (1,726 pairs)			
Benchmark	RR	ΔRR	CE ratio	Live	RR	ΔRR	CE ratio	Live
Ours: CAPS-fast (progressive voting); recall 69.3 / 29.2 %, 292 / 505 k CE								
RANSAC, classical stopping (p=0.999)	69.5	−0.1 (p=0.85)	0.84×	1.29×	26.3	+2.8 (p=1.3 × 10^{-5})	0.76×	1.21×
RANSAC, 100k hypotheses	69.5	−0.2 (p=0.92)	3.48×	6.03×	26.5	+2.6 (p=1.6 × 10^{-5})	0.91×	1.46×
SC^2 (simplified, same harness)	71.8	−2.5	–	1.86×	27.6	+1.5	–	0.81×
Largest search config., run once	66.7	+2.6	1.66×	1.47×	27.9	+1.3	0.84×	0.74×
Anytime schedule run to completion	69.7	−0.3 (p=0.77)	2.88×	–	29.3	−0.2 (p=0.77)	1.45×	–
Ours: CAPS-accurate (full voting); recall 71.2 / 31.2 %, 749 / 1,511 k CE								
RANSAC, classical stopping (p=0.999)	69.5	+1.7 (p=0.08)	0.33×	0.40×	26.3	+4.8 (p=9.5 × 10^{-12})	0.26×	0.30×
RANSAC, 100k hypotheses	69.5	+1.7 (p=0.06)	1.36×	3.33×	26.5	+4.6 (p=6.3 × 10^{-12})	0.30×	0.34×
SC^2 (simplified, same harness)	71.8	−0.7	–	0.66×	27.6	+3.5	–	0.21×
Largest search config., run once	69.3	+1.8	2.14×	2.31×	31.0	+0.2	0.93×	0.82×
Anytime schedule run to completion	71.4	−0.2 (p=1.00)	3.48×	–	31.0	+0.2 (p=0.55)	1.51×	–

Table 18. **Certified controllers for CAPS** (test). CE: mean cost (k); UB: upper bound of the recall loss against the schedule run to completion (pp; ✓ non-inferior); Fr.: recall above the static frontier of the same variant at equal cost (pp).

	3DMatch				3DLoMatch			
Variant, policy	CE	RR	UB	Fr.	CE	RR	UB	Fr.
full, threshold	1,051	70.6	1.9 ✗	−0.1	1,906	31.0	0.2 ✓	+0.0
full, give-up	1,033	69.9	2.5 ✗	−0.7	1,668	30.0	1.5 ✓	−1.0
full, Markov	**749**	71.2	0.8 ✓	+0.6	**1,511**	31.2	0.1 ✓	+0.2
progr., threshold	357	68.6	2.1 ✗	−0.3	607	29.5	0.0 ✓	+1.6
progr., give-up	359	68.6	2.1 ✗	−0.3	585	29.4	0.2 ✓	+1.5
progr., Markov	**292**	69.3	0.9 ✓	+0.4	**505**	29.2	0.8 ✓	+1.3

Table 19. **Where RANSAC spends its budget** under the classical stopping rule, by ground-truth inliers among the 1,000 correspondences: pairs, share of the evaluation budget (%), recall (%).

	3DMatch			3DLoMatch		
GT inliers	pairs	budget	recall	pairs	budget	recall
[0,5)	80	10	0.0	569	34	0.4
[5,10)	104	13	2.9	370	24	4.1
[10,20)	188	26	31.9	384	24	32.6
[20,40)	250	32	78.4	266	15	73.3
[40,80)	298	15	97.7	107	3	98.1
[80,∞)	359	4	99.7	30	0	100.0

Table 20. **Evaluation pitfalls for adaptive registration**, each observed in this study.

Pitfall	What happened	Rule
Weak fixed baseline	The pose-search schedule run to completion re-votes at every grid density; against the largest configuration run once, the 3DLoMatch saving of CAPS turns from 1.5× into 0.84–0.93×.	Compare with the strongest configuration run once and with the static frontier.
Misattributed timing gaps	A validation-vs-test gap in full-budget time (1,672 ms vs. 2,762 ms for SC^2-PCR) looked like machine load; it was content (validation pairs have fewer correspondences). Re-timing reproduced the logs within a few percent.	Compare like with like; time a method with its reference in the same run.
Microbenchmarks	The ordering costs 7.2 ms warm and alone, 29 ms per pair inside the parallel logs.	Charge in-pipeline costs; let live timing decide.
Reference drift	For ladders that stop below the full budget, risk was first measured against the ladder's own last stage, and the fallback was assumed safe.	Certify against the full budget, and certify the fallback.
Average-case certificates	A $\leq 1\,\%$ marginal risk certified on the odometry pairs became 1.37 % on the harder KITTI raw drives (TurboReg, ratio order).	Certify per difficulty group.
Fitting data without failures	FCGF's outdoor controller was fitted on FCGF's own training pairs, where nearly every first stage succeeds; it learned no failure signal, and the certificate kept the full budget although 250 correspondences register 92.1 % of the test pairs.	Fit on data the descriptor has not been trained on.
Reused test sets	The official test sets had been evaluated by earlier versions of this work, and the 3DMatch validation scenes served both design and calibration.	Pre-register a protocol with fresh blind test sets; report reused ones as development benchmarks.
Averages hide decisive pairs	Tripling the mean number of inliers in the cheap budget (one-pass anchors) gave almost no success gain.	Judge design choices on pairs near the decision boundary.
One cost metric	CAPS does more CE than RANSAC yet is faster; TurboReg's operation count hides a 4.3× cost increase on consistent input.	Report hardware-independent and wall-clock cost.
Free repeated stages	Pairs with $\leq 2{,}000$ correspondences got their full budget for free when a ladder skipped the stage holding the real run.	Charge each distinct computation once; unit-test the accounting.
Implementation overhead	The gate costs 3 ms when timed alone but 40–48 ms per pair inside the live pipeline, enough to turn a gain into a loss for SC^2-PCR above 30 % overlap.	Time every component inside the pipeline it serves, never only alone.
Improvements found after a blind test	The gate and the classifier choice were found after the protocol-v2 blind evaluation; on fresh pairs they held for TurboReg and at low overlap, not for SC^2-PCR above 30 % overlap.	Pre-register and test again on data nothing has processed.
Selection after certification	We first certified every candidate ladder on the calibration pairs and kept the cheapest, which voids the guarantee (it then holds only by a union bound over the candidates); choosing on the fitting pairs lowered our 3DMatch speed-ups by about a tenth.	Choose on data the certificate never sees; certify only the choice.

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